

Q1 2021 (01 Sep Shift 2)

The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point $(2, -1, 6)$ is :

- (1) $\sqrt{26}$
- (2) $2\sqrt{5}$
- (3) $2\sqrt{6}$
- (4) $4\sqrt{2}$

Q2 2021 (01 Sep Shift 2)

Let the acute angle bisector of the two planes $x - 2y - 2z + 1 = 0$ and $2x - 3y - 6z + 1 = 0$ be the plane P. Then which of the following points lies on P?

- (1) $(3, 1, -\frac{1}{2})$
- (2) $(-2, 0, -\frac{1}{2})$
- (3) $(0, 2, -4)$
- (4) $(4, 0, -2)$

Q3 2021 (31 Aug Shift 2)

Suppose the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. Then $(\alpha + \beta)$ is equal to .

Q4 2021 (31 Aug Shift 2)

The distance of the point $(-1, 2, -2)$ from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is :

- (1) $\frac{1}{\sqrt{2}}$
- (2) $\frac{5}{2}$
- (3) $\frac{\sqrt{42}}{2}$
- (4) $\frac{\sqrt{34}}{2}$

Q5 2021 (31 Aug Shift 1)

The square of the distance of the point of intersection of the line $\frac{x+1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is .

Q6 2021 (31 Aug Shift 1)

Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and

$x + 5y - 2z + 9 = 0$, be $\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is equal to :

(1) -23

(2) -15

(3) 23

(4) 15

Q7 2021 (27 Aug Shift 2)

Let S be the mirror image of the point $Q(1, 3, 4)$ with respect to the plane $2x - y + z + 3 = 0$ and let $R(3, 5, \gamma)$ be a point of this plane. Then the square of the length of the line segment SR is .

Q8 2021 (27 Aug Shift 2)

The equation of the plane passing through the line of intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0$ and parallel to the x-axis is:

(1) $\vec{r} \cdot (\hat{j} - 3\hat{k}) + 6 = 0$

(2) $\vec{r} \cdot (\hat{i} + 3\hat{k}) + 6 = 0$

(3) $\vec{r} \cdot (\hat{i} - 3\hat{k}) + 6 = 0$

(4) $\vec{r} \cdot (\hat{j} - 3\hat{k}) - 6 = 0$

Q9 2021 (27 Aug Shift 2)

The angle between the straight lines, whose direction cosines are given by the equations $2l + 2m - n = 0$ and $mn + nl + lm = 0$, is :

- (1) $\frac{\pi}{2}$
- (2) $\pi - \cos^{-1}\left(\frac{4}{9}\right)$
- (3) $\cos^{-1}\left(\frac{8}{9}\right)$
- (4) $\frac{\pi}{3}$

Q10 2021 (27 Aug Shift 1)

Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$ is :

- (1) $3x - y - 5z + 2 = 0$
- (2) $3x - 4z + 3 = 0$
- (3) $-x + 2y + 2z - 3 = 0$
- (4) $4x - y - 5z + 2 = 0$

Q11 2021 (27 Aug Shift 1)

The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are $2, 3, -6$ is :

- (1) 3
- (2) 5
- (3) 2
- (4) 1

Q12 2021 (26 Aug Shift 2)

Let Q be the foot of the perpendicular from the point $P(7, -2, 13)$ on the plane containing the lines

$\frac{x+1}{6} = \frac{y-1}{7} = \frac{z-3}{8}$ and $\frac{x-1}{3} = \frac{y-2}{5} = \frac{z-3}{7}$. Then $(PQ)^2$, is equal to .

Q13 2021 (26 Aug Shift 2)

Let P be the plane passing through the point $(1, 2, 3)$ and the line of intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + 4\hat{k}) = 16$ and $\vec{r} \cdot (-\hat{i} + \hat{j} + \hat{k}) = 6$. Then which of the following points does NOT lie on P ?

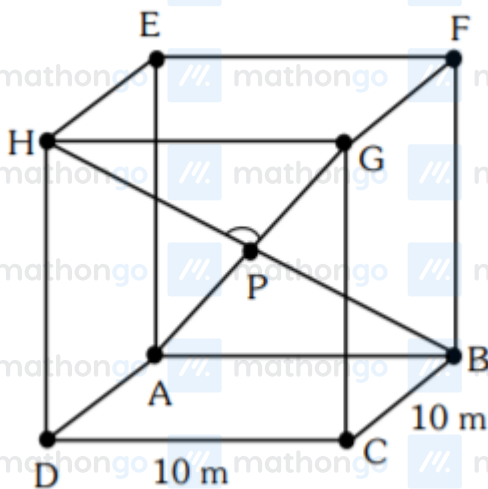
- (1) $(3, 3, 2)$
- (2) $(6, -6, 2)$
- (3) $(4, 2, 2)$
- (4) $(-8, 8, 6)$

Q14 2021 (26 Aug Shift 2)

A hall has a square floor of dimension $10 \text{ m} \times 10 \text{ m}$ (see the figure) and vertical walls.

If the angle GPH between the diagonals AG and BH is

$\cos^{-1} \frac{1}{5}$, then the height of the hall (in meters) is :



- (1) 5
- (2) $2\sqrt{10}$
- (3) $5\sqrt{3}$
- (4) $5\sqrt{2}$

Q15 2021 (26 Aug Shift 1)

Let the line L be the projection of the line

$$\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$$

in the plane $x - 2y - z = 3$. If d is the distance of the point $(0, 0, 6)$ from L , then d^2 is equal to .

Q16 2021 (26 Aug Shift 1)

A plane P contains the line

$$x + 2y + 3z + 1 = 0 = x - y - z - 6,$$

and is perpendicular to the plane $-2x + y + z + 8 = 0$.

Then which of the following points lies on P ?

(1) $(-1, 1, 2)$

(2) $(0, 1, 1)$

(3) $(1, 0, 1)$

(4) $(2, -1, 1)$

Answer Key

Q1 (3)

Q2 (2)

Q3 (7)

Q4 (4)

Q5 (61)

Q6 (1)

Q7 (72)

Q8 (1)

Q9 (1)

Q10 (4)

Q11 (4)

Q12 (96)

Q13 (3)

Q14 (4)

Q15 (26)

Q16 (2)

Q1 (3)

$$3y - 2z - 1 = 0 = 3x - z + 4$$

$$3y - 2z - 1 = 0 \text{ D.R's } \Rightarrow (0, 3, -2)$$

$$3x - z + 4 = 0 \text{ D.R's } \Rightarrow (3, -1, 0)$$

Let DR's of given line are a, b, c

$$\text{Now } 3b - 2c = 0 \text{ \& } 3a - c = 0$$

$$\therefore 6a = 3b = 2c$$

$$a : b : c = 3 : 6 : 9$$

Any pt on line

$$3K - 1, 6K + 1, 9K + 1$$

$$\text{Now } 3(3K - 1) + 6(6K + 1) + 9(9K + 1) = 0$$

$$\Rightarrow K = \frac{1}{3}$$

$$\text{Point on line } \Rightarrow (0, 3, 4)$$

$$\text{Given point } (2, -1, 6)$$

$$\Rightarrow \text{Distance} = \sqrt{4 + 16 + 4} = 2\sqrt{6}$$

Option (3)

Q2 (2)

$$P_1 : x - 2y - 2z + 1 = 0$$

$$P_2 : 2x - 3y - 6z + 1 = 0$$

$$\left| \frac{x-2y-2z+1}{\sqrt{1+4+4}} \right| = \left| \frac{2x-3y-6z+1}{\sqrt{2^2+3^2+6^2}} \right|$$

$$\frac{x-2y-2z+1}{3} = \pm \frac{2x-3y-6z+1}{7}$$

$$\text{Since } a_1a_2 + b_1b_2 + c_1c_2 = 20 > 0$$

$\therefore$ Negative sign will give acute bisector

$$7x - 14y - 14z + 7 = -[6x - 9y - 18z + 3]$$

$$\Rightarrow 13x - 23y - 32z + 10 = 0$$

$$\left(-2, 0, -\frac{1}{2}\right) \text{ satisfy it } \therefore \text{Ans (2)}$$

Q3 (7)

Point $(2, 2, -2)$ also lies on given plane

$$\text{So } 2 + 3 \times 2 - 2(-2) + \beta = 0$$

$$\Rightarrow 2 + 6 + 4 + \beta = 0 \Rightarrow \beta = -12$$

$$\text{Also } \alpha \times 1 - 5 \times 3 + 2 \times -2 = 0$$

$$\Rightarrow \alpha - 15 - 4 = 0 \Rightarrow \alpha = 19$$

$$\therefore \alpha + \beta = 19 - 12 = 7$$

Q4 (4)

$$P_1 : 2x + 3y + 2z = 0$$

$$\Rightarrow \vec{n}_1 = 2\hat{i} + 3\hat{j} + 2\hat{k}$$

$$P_2 : x - 2y + z = 0$$

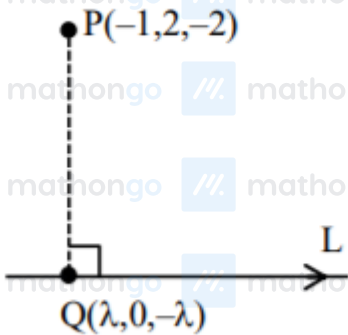
$$\Rightarrow \vec{n}_2 = \hat{i} - 2\hat{j} + \hat{k}$$

Direction vector of line L which is line of intersection of P_1 & P_2

$$\vec{r} = \vec{n}_1 \times \vec{n}_2 = 7\hat{i} - 7\hat{k}$$

DR's of L are $(1, 0, -1)$

$$\Rightarrow \text{Equation of L : } \frac{x}{1} = \frac{y}{0} = \frac{z}{-1} = \lambda$$



$$\text{DR's of } \overrightarrow{PQ} = (\lambda + 1, -2, 2 - \lambda)$$

$$\because \overrightarrow{PQ} \perp \vec{r}$$

$$\Rightarrow (\lambda + 1)(1) + (-2)(0) + (2 - \lambda)(-1) = 0$$

$$\Rightarrow \lambda = \frac{1}{2} \Rightarrow Q\left(\frac{1}{2}, 0, \frac{-1}{2}\right)$$

$$\Rightarrow PQ = \frac{\sqrt{34}}{2}$$

Q5 (61)

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6} = \lambda$$

$$x = 2\lambda + 1, y = 3\lambda + 2, z = 6\lambda - 1$$

for point of intersection of line & plane

$$2(2\lambda + 1) - (3\lambda + 2) + (6\lambda - 1) = 6$$

$$7\lambda = 7 \Rightarrow \lambda = 1$$

point : (3, 5, 5)

$$(\text{distance})^2 = (3+1)^2 + (5+1)^2 + (5-2)^2$$

$$= 16 + 36 + 9 = 61$$

Q6 (1)

Equation of plane is

$$3x - 2y + 4z - 7 + \lambda(x + 5y - 2z + 9) = 0$$

$$(3 + \lambda)x + (5\lambda - 2)y + (4 - 2\lambda)z + 9\lambda - 7 = 0$$

passing through (1, 4, -3)

$$\Rightarrow 3 + \lambda + 20\lambda - 8 - 12 + 6\lambda + 9\lambda - 7 = 0$$

$$\Rightarrow \lambda = \frac{2}{3}$$

 $\Rightarrow$ equation of plane is

$$-11x - 4y + 8z + 3 = 0$$

$$\Rightarrow \alpha + \beta + \gamma = -23$$

Q7 (72)Since R(3, 5, γ) lies on the plane $2x - y + z + 3 = 0$.

$$\text{Therefore, } 6 - 5 + \gamma + 3 = 0$$

$$\Rightarrow \gamma = -4$$

Now,

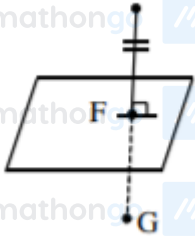
dr's of line QS

are $2, -1, 1$

equation of line QS is

$$\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = \lambda \text{ (say)}$$

$$\Rightarrow F(2\lambda + 1, -\lambda + 3, \lambda + 4)$$



F lies in the plane

$$\Rightarrow 2(2\lambda + 1) - (-\lambda + 3) + (\lambda + 4) + 3 = 0$$

$$\Rightarrow 4\lambda + 2 + \lambda - 3 + \lambda + 7 = 0$$

$$\Rightarrow 6\lambda + 6 = 0 \Rightarrow \lambda = -1$$

$$\Rightarrow F(-1, 4, 3)$$

Since, F is mid-point of QS .

Therefore, co-ordinates of S are $(-3, 5, 2)$

$$\text{So, } SR = \sqrt{36 + 0 + 36} = \sqrt{72}$$

$$SR^2 = 72.$$

Q8 (1)

Equation of planes are

$$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) - 1 = 0 \Rightarrow x + y + z - 1 = 0$$

$$\text{and } \vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) + 4 = 0 \Rightarrow 2x + 3y - z + 4 = 0$$

equation of planes through line of intersection of these planes is :-

$$(x + y + z - 1) + \lambda(2x + 3y - z + 4) = 0$$

$$\Rightarrow (1 + 2\lambda)x + (1 + 3\lambda)y + (1 - \lambda)z - 1 + 4\lambda = 0$$

But this plane is parallel to x -axis whose direction are $(1, 0, 0)$

$$\therefore (1 + 2\lambda)1 + (1 + 3\lambda)0 + (1 - \lambda)0 = 0$$

$$\lambda = -\frac{1}{2}$$

$\therefore$ Required plane is

$$0x + \left(1 - \frac{3}{2}\right)y + \left(1 + \frac{1}{2}\right)z - 1 + 4\left(\frac{-1}{2}\right) = 0$$

$$\Rightarrow \frac{-y}{2} + \frac{3}{2}z - 3 = 0$$

$$\Rightarrow y - 3z + 6 = 0$$

$$\Rightarrow \vec{r} \cdot (\hat{j} - 3\hat{k}) + 6 = 0 \text{ Ans.}$$

Q9 (1)

$$n = 2(\ell + m)$$

$$\ell m + n(\ell + m) = 0$$

$$\ell m + 2(\ell + m)^2 = 0$$

$$2\ell^2 + 2m^2 + 5\ell m = 0$$

$$2\left(\frac{\ell}{m}\right)^2 + 2 + 5\left(\frac{\ell}{m}\right) = 0$$

$$2t^2 + 5t + 2 = 0$$

$$(t + 2)(2t + 1) = 0$$

$$\Rightarrow t = -2; -\frac{1}{2}$$

$$(i) \frac{\ell}{m} = -2$$

$$\frac{n}{m} = -2$$

$$(-2m, m, -2m)$$

$$(-2, 1, -2)$$

$$(ii) \frac{\ell}{m} = -\frac{1}{2}$$

$$n = -2\ell$$

$$(\ell, -2\ell, -2\ell)$$

$$(1, -2, -2)$$

$$\cos \theta = \frac{-2-2+4}{\sqrt{9}\sqrt{9}} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

Q10 (4)

Required equation of plane

$$P_1 + \lambda P_2 = 0$$

$$(x - y - z - 1) + \lambda(2x + y - 3z + 4) = 0$$

Given that its dist. From origin is $\frac{2}{\sqrt{21}}$

$$\text{Thus } \frac{|4\lambda - 1|}{\sqrt{(2\lambda+1)^2 + (\lambda-1)^2 + (-3\lambda-1)^2}} = \frac{\sqrt{2}}{\sqrt{21}}$$

$$\Rightarrow 21(4\lambda - 1)^2 = 2(14\lambda^2 + 8\lambda + 3)$$

$$\Rightarrow 336\lambda^2 - 168\lambda + 21 = 28\lambda^2 + 16\lambda + 6$$

$$\Rightarrow 308\lambda^2 - 184\lambda + 15 = 0$$

$$\Rightarrow 308\lambda^2 - 154\lambda - 30\lambda + 15 = 0$$

$$\Rightarrow (2\lambda - 1)(154\lambda - 15) = 0$$

$$\Rightarrow \lambda = \frac{1}{2} \text{ or } \frac{15}{154}$$

for $\lambda = \frac{1}{2}$ reqd. plane is

$$4x - y - 5z + 2 = 0$$

Q11 (4)

A(1, -2, 3)

$$\vec{r} = (1, -2, 3) + \lambda(2, 3 - 6)$$

$$(1 + 2\lambda, -2 + 3\lambda, 3 - 6\lambda)$$

$$x - y + z = 5$$

$$(1 + 2\lambda) + 2 - 3\lambda + 3 - 6\lambda = 5$$

$$\Rightarrow 6 - 7\lambda = 5 \Rightarrow \lambda = \frac{1}{7}$$

$$\text{so, } P = \left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$$

$$AP = \sqrt{\left(1 - \frac{9}{7}\right)^2 + \left(-2 + \frac{11}{7}\right)^2 + \left(3 - \frac{15}{7}\right)^2}$$

$$AP = \sqrt{\left(\frac{4}{7}\right)^2 + \frac{9}{49} + \frac{36}{49}} = 1$$

Q12 (96)

Containing the line $\begin{vmatrix} x+1 & y-1 & z-3 \\ 6 & 7 & 8 \\ 3 & 5 & 7 \end{vmatrix} = 0$

$$9(x+1) - 18(y-1) + 9(z-3) = 0$$

$$x - 2y + z = 0$$

$$PQ = \left| \frac{7+4+13}{\sqrt{6}} \right| = 4\sqrt{6}$$

$$PQ^2 = 96$$

Q13 (3)

$$(x + y + 4z - 16) + \lambda(-x + y + z - 6) = 0$$

Passes through (1, 2, 3)

$$-1 + \lambda(-2) \Rightarrow \lambda = -\frac{1}{2}$$

$$2(x + y + 4z - 16) - (-x + y + z - 6) = 0$$

$$3x + y + 7z - 26 = 0$$

Q14 (4)

$$A(\hat{j}) \cdot B(10\hat{i})$$

$$H(\hat{h} + 10\hat{k})$$

$$G(10\hat{i} + h\hat{j} + 10\hat{k})$$

$$\overrightarrow{AG} = 10\hat{i} + h\hat{j} + 10\hat{k}$$

$$\overrightarrow{BH} = -10\hat{i} + h\hat{j} + 10\hat{k}$$

$$\cos \theta = \frac{\overrightarrow{AG} \cdot \overrightarrow{BH}}{|\overrightarrow{AG}| |\overrightarrow{BH}|}$$

$$\frac{1}{5} = \frac{h^2}{h^2 + 200}$$

$$4h^2 = 200 \Rightarrow h = 5\sqrt{2}$$

Q15 (26)

$$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$$

for foot of $\perp$ r of (1, 3, 4) on $x - 2y - z - 3 = 0$

$$(1+t) - 2(3-2t) - (4-t) - 3 = 0$$

$$\Rightarrow t = 2$$

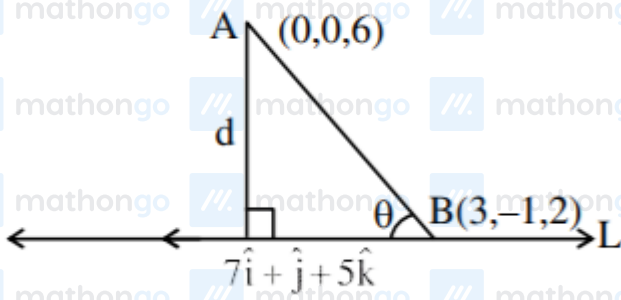
$$\text{So foot of } \perp r \triangleq (3, -1, 2)$$

& point of intersection of L_1 with plane

$$\text{is } (-11, -3, -8)$$

$$\text{dr's of } L \text{ is } \langle 14, 2, 10 \rangle$$

$$\cong \langle 7, 1, 5 \rangle$$

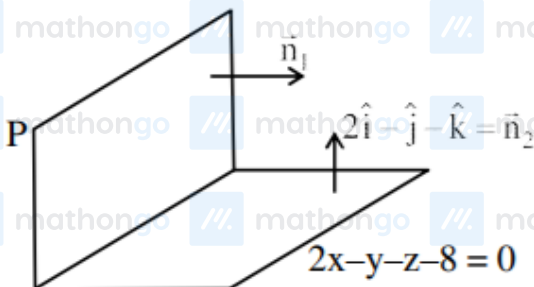


$$d = AB \sin \theta = \frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -4 \\ 7 & 1 & 5 \end{vmatrix}}{\sqrt{7^2 + 1^2 + 5^2}}$$

$$\Rightarrow d^2 = \frac{1^2 + (43)^2 + (10)^2}{49 + 1 + 25} = 26$$

Q16 (2)

Equation of plane P can be assumed as



$$P: x + 2y + 3z + 1 + \lambda(x - y - z - 6) = 0$$

$$\Rightarrow P : (1 + \lambda)x + (2 - \lambda)y + (3 - \lambda)z + 1 - 6\lambda = 0$$

$$\Rightarrow \vec{n}_1 = (1 + \lambda)\hat{i} + (2 - \lambda)\hat{j} + (3 - \lambda)\hat{k}$$

$$\therefore \vec{n}_1 \cdot \vec{n}_2 = 0$$

$$\Rightarrow 2(1 + \lambda) - (2 - \lambda) - (3 - \lambda) = 0$$

$$\Rightarrow 2 + 2\lambda - 2 + \lambda - 3 + \lambda = 0 \Rightarrow \lambda = \frac{3}{4}$$

$$\Rightarrow P : \frac{7x}{4} + \frac{5y}{4} + \frac{9z}{4} - \frac{14}{4} = 0$$

$$\Rightarrow 7x + 5y + 9z = 14$$

(0, 1, 1) lies on P